

Summary

Computer Science and Cyber Operations researcher and student with two published IEEE papers spanning post-quantum cryptography and quantum-classical machine learning. Experienced in building software from scratch in Zig, C/C++, Python, and TypeScript, with a focus on systems programming, developer tooling, and applied quantum/ML research.

Education

Madison, SD **Dakota State University** **Fall 2023 – Spring 2027**

- B.S. Computer Science and Cyber Operations, GPA: 3.96
- Dean's List, Honors Program, DSU Presidential Champion Scholarship, Kappa Sigma Iota Honor Society

Work Experience

Quantum Security Researcher **Dakota State University** **Summer 2024 – Spring 2025**

- Published paper:* Multilevel Post Quantum Encryption for Images with Quantum Fourier Transform (IEEE eIT)
- Designed a novel multilevel post-quantum image encryption framework (QFT + AES + LWE) for medical, defense, and financial images.
 - Preserved >40 dB PSNR and >0.95 SSIM quality with encryption processing times under 2.4 seconds.

Quantum AI Researcher **Dakota State University** **Fall 2025 – Present**

- Published paper:* A Novel Hybrid Quantum-Classical Path Optimization for Methane Detection Using Remote Quantum Intensity Prediction Models (IEEE Access)
- Collaborated with Savannah River National Laboratory to develop a hybrid quantum-classical drone routing system (CatBoost + QAOA) built on SRNL provided RAMS/HYSPLIT dataset, cutting methane leak detection energy use by 50% versus classical baselines.
 - Implemented a custom Quantum Gaussian Process Regression model (QGPR) in Python, achieving a 20.67% R^2 improvement over classical GPR.

STEM Academic Tutor **Dakota State University** **Fall 2024 – Spring 2025**

- Mentored GrowStem scholarship recipients in mathematics and programming fundamentals.
- Guided students through initial phases of undergraduate research project development.

Technical Experience

Projects

- **Shellu:** Built a POSIX-compatible shell from scratch in Zig, implementing pipelines, I/O redirection, background job control, command history, tab completion, and builtin commands using POSIX process APIs, system calls, and C interoperability.
- **Tracedog:** Built a TypeScript observability tool that ingests OpenTelemetry traces and uses LLM assisted causal analysis to automatically generate root cause reports, reducing manual trace correlation during incident response.
- **ZNN:** A neural network library implemented in Zig. It includes forward propagation, back propagation, and gradient descent, without third party ML frameworks.
- **Zig Contributor:** Contribute to the zig programming language through open source development.

Languages and Technologies

- **Languages:** Zig, C, C++, Python, SQL, Java, C#, HTML, CSS, JavaScript, TypeScript
- **Technologies:** Git, GitHub, GitHub Actions, SQLite, PostgreSQL, Svelte, SvelteKit, Docker, PyTorch, Qiskit